Xu Han

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Education

Munich University of Applied Sciences *Mar 2026 - Present*
M.Sc. in Photonics
- **Selected coursework:** Physical Optics and Modelling (Zemax); Micro and Fiber Optics; Optical Measurement Technology; Laser Physics - Non-linear Optics - Ultrafast Optics; Semiconductor Optics and Components.

Ludwig Maximilian University of Munich (LMU) *Apr 2022 - Aug 2024*
M.Sc. in Astrophysics
- Grade: 1.55 (German scale)
- **Selected coursework:** Essentials of Advanced Astrophysics; Cosmology and Structure Formation; Statistical Methods; Advanced Astrophysical Seminar; Instrumental Laboratory Courses.

University of Duisburg-Essen *Oct 2018 - Mar 2022*
B.Sc. in Physics
- Grade: 1.8 (German scale)
- **Selected coursework:** Quantum Mechanics; Statistical Mechanics; Fundamentals of Astrophysics; Advanced Laboratory Courses.

Hubei University of Technology *Sep 2013 - Jul 2017*
B.Eng. in Electrical Engineering and Automation
- Training in electronics, signals and systems, control systems and embedded systems.

Research Experience

Master's thesis research | LMU Munich *Mar 2023 - Aug 2024*
Thesis: Watershed Cosmic Void Finder for Weak Gravitational Lensing Voids on 2D Full-Sky Convergence Maps
Supervisors: Prof. Joseph Mohr and Dr. Chris Davies.
- Developed a watershed void finder for 2D full-sky weak-lensing convergence maps; assessed how algorithm parameters affect the recovered void populations.
- Characterised void abundance, ellipticity and convergence profiles; adapted the algorithm to DES Y3 weak-lensing maps.
- Explored cosmological forecasting with weak-lensing void statistics using machine learning and MCMC.

Bachelor's thesis research | University of Duisburg-Essen *Mar 2021 - Aug 2021*
Thesis: Design of an Experimental Setup for the Measurement of Collisional Charging on Granulate.
Supervisors: Prof. Gerhard Wurm and Dr. Felix Jungmann.
- Designed the apparatus in CAD, 3D-printed and assembled its mechanical components, and integrated and debugged the electronic charge-measurement modules.
- Performed experiments and data analysis to assess grain-size dependence of collisional charging and test a hypothesis about dust growth during early planet formation.

Laboratory and Observational Experience

- **Photonics:** APD measurements, laser spectra, nonlinear optics, laser interferometry, fiber Bragg gratings and fiber coupling.
- **Astronomical instrumentation:** CCD imaging and data reduction, optical interferometry, deformable-mirror adaptive-optics tuning, and radio measurements of Galactic rotation.
- **Observing:** Participated in observations and image processing at Wendelstein Observatory using the 43 cm telescope and 2 m Fraunhofer telescope.

Selected Independent Projects

Adaptive Optics - Basics

- Built a Python single-conjugate AO model from first principles: von Kármán turbulence, Shack-Hartmann sensing, TSVD wavefront reconstruction, deformable-mirror correction and closed-loop control with delay.
- Validated turbulence statistics, reconstruction noise, loop stability and J/H/K PSFs against analytic theory; developed seven tutorial notebooks, including an exploratory Fourier-optics pyramid sensor.

Astrophotonics

- Modelled how AO wavefront error, injection jitter and photonic-lantern mode count affect fiber coupling, spectral SNR and radial-velocity precision; validated propagation and energy conservation.

AO-Fed IFU Spectrograph Simulation

- Modelled the source-to-detector chain and cube reconstruction of an AO-fed IFU spectrograph; compared flux, velocity and SNR with synthetic truth and examined calibration-error sensitivity.

Optoelectronic Signal Chain Simulations

- Developed Python case studies of photodetector/APD noise, laser thermal control, LiDAR detection, camera sensors and spectrometer calibration, with parameter scans and uncertainty analysis.

Work Experience

Python Developer | BaryonBib *Dec 2024 - Feb 2025*
MAX!mize start-up team, MPA, Garching

- Contributed to a centralised bibliography platform in Python; implemented and tested LLM-based functionality.

Talks, Workshops and Schools

MAVIS 2026 workshop *Sep 2026*

Tonale Winter School on Cosmology *Dec 2024*

ORIGINS Lensing Day *Nov 2024*
Watershed on the curved sky

Cambridge-LMU Cosmology Meeting *Dec 2023*

Advanced Astrophysical Seminar | LMU Munich *Nov 2022*
40-minute course presentation and literature study on large ground-based astronomical telescopes, telescope design types and adaptive optics.

Skills

Programming and data analysis: Python (NumPy, SciPy, Matplotlib, pandas, scikit-learn, TensorFlow); MATLAB; Mathematica; statistical analysis and MCMC.

Optical modelling: Wavefront, PSF and fiber-coupling simulations; TIPTOP; Zemax (course-based training).

Experimental methods: CAD, 3D printing, mechanical assembly, electronic measurement integration, optical alignment, imaging and interferometry.

Research tools: Linux, Git and LaTeX.

Languages: Chinese (native); English (fluent); German (fluent); French (beginner).

Hobbies

Observational astronomy, astrophotography and mountaineering.

References available on request.